Classical/Quantum Boundary Diagnostic

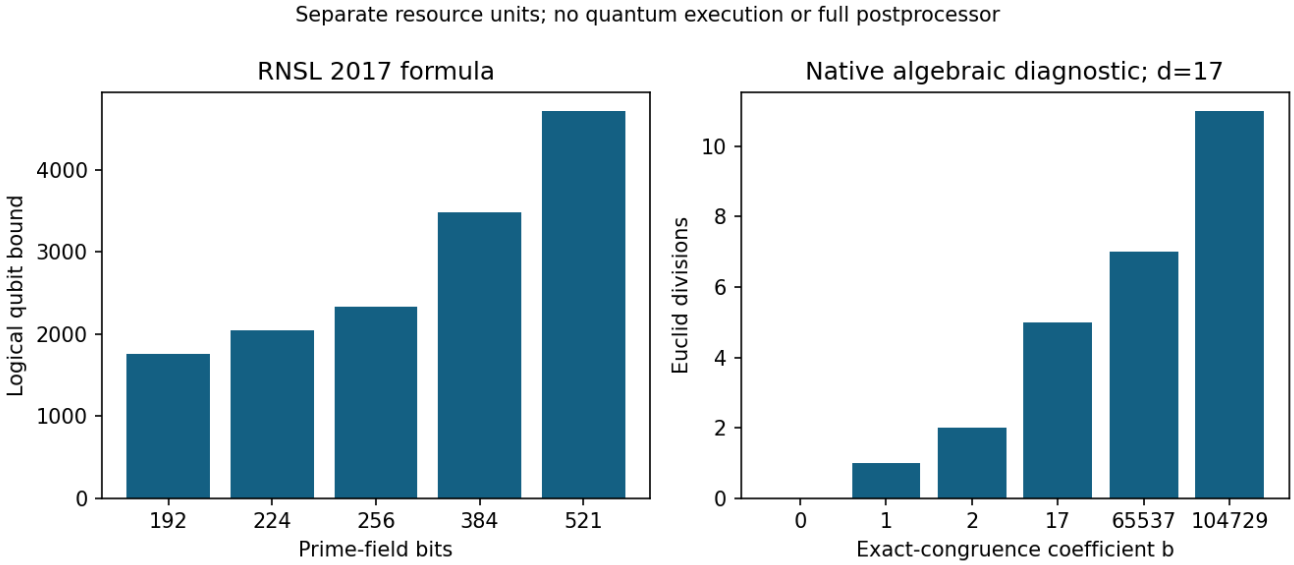
Research execution record

18 October 2026

Partial. No quantum execution or full Shor postprocessor.

The native program emitted 75 resource rows and checked 18 exact-congruence inputs: 15 recovered the chosen integer and three noninvertible zero coefficients were rejected. Unit tests additionally cover every secret and unit modulo 101 and a composite nonunit.

Resource rows keep public-point doublings, coordinate bytes, logical qubits and Toffoli bounds separate. At 256 prime-field bits the RNSL 2017 formula gives 2,330 logical qubits and approximately 1.287×10^{11} Toffoli gates per shot. Repeated-shot probabilities are explicitly assumed sensitivity inputs. They are not measured circuit success probabilities.



The algebraic input $a + bd = 0 \pmod{r}$ is exact and synthetic. The experiment does not recover this congruence from noisy power-of-two Fourier samples. It does not time actual point tables, verify an elliptic-curve candidate, simulate QFT/feed-forward, or estimate fault-tolerant physical resources. These are remaining obligations for item 20.

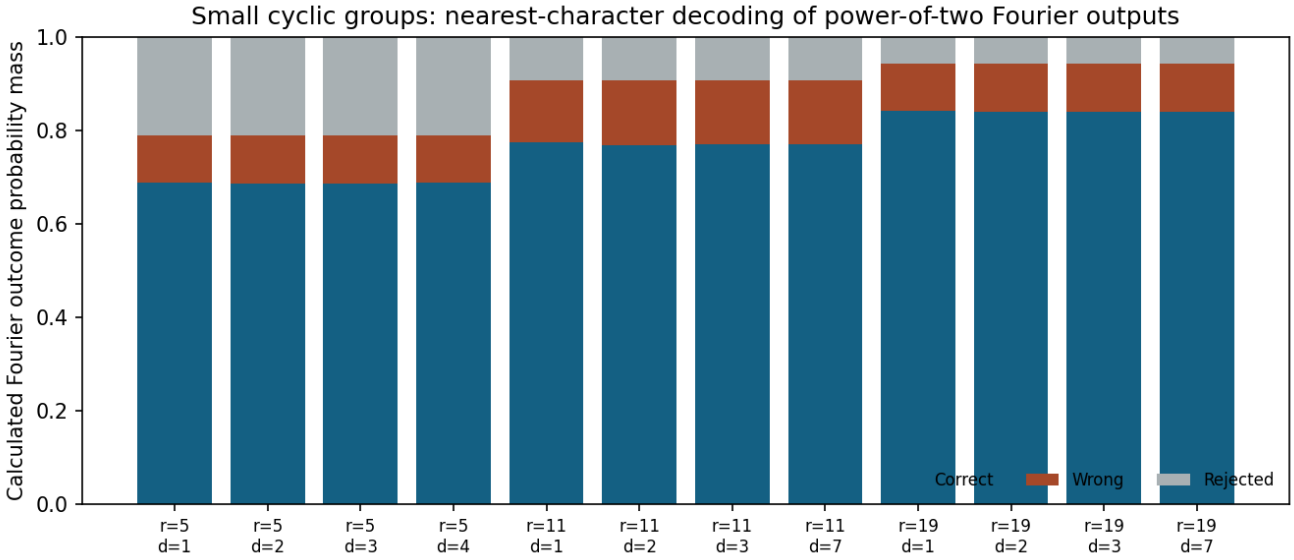
Evidence. The frozen protocol, Rust source, 75 resource rows, 18 diagnostics, seven-test output and command/source manifest are retained beside this report. Execution revision: `b3a3321c`.

Sources. Roetteler et al., arXiv:1706.06752 (2017); Häner et al., arXiv:2001.09580 (2020), section 2. No gate counts from different architectures are combined.

Finite Fourier-output follow-up

A separate frozen protocol calculates the actual power-of-two Fourier output law on twelve synthetic cyclic groups: orders 5, 11 and 19, using transform sizes 16, 32 and 64. The native code traces out the group register and Fourier transforms every coset indicator. All laws normalize within 10^{-10}.

The nearest-character decoder retains correct, incorrect and rejected candidate masses. It is a heuristic with observed failures, not a complete recovery algorithm. No small-group probability is transferred to deployed parameters.



Three additional tests pass, including comparison of the FFT with a naive DFT and symmetry of the $d = 1$ outcome law. The standalone crate now has ten passing tests. This is classical numerical simulation of a small cyclic-group distribution. Candidate checking uses the known synthetic integer and does not execute elliptic-curve verification or a quantum circuit.

Evidence. Separate Fourier protocol, native source and all twelve complete probability tables are retained in the campaign directory. Source revision `b7528cb5`; binary `fourier-recovery`.